

GURUPRASAD VISWANATHAN RAMESH

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EDUCATION

Jan 2024 – Present	Ph.D., Electrical and Computer Engineering University of Wisconsin – Madison	GPA: 3.88/4 <i>Expected Dec 2027</i>
Sep 2021 – Dec 2023	M.S. (Research), Electrical and Computer Engineering University of Wisconsin – Madison	
Aug 2016 – Jun 2020	B.Tech., Electronics and Communication Engineering National Institute of Technology, Tiruchirappalli, India Thesis: <i>Dimension Estimation from Depth Map of Monocular Images</i>	GPA: 8.34/10

RESEARCH INTERESTS

Trustworthy AI, Fairness & Auditing, ML Security and Privacy, LLMs, Web Agents, ASR, Evaluation & Benchmarking

PUBLICATIONS

PETS '26	WebSP-Eval: Evaluating Web Agents on Website Security and Privacy Tasks [Code] [Paper] [Project] Guruprasad V. Ramesh , Asmit Nayak, Basieem Siddique, Kassem Fawaz. <i>Proceedings on Privacy Enhancing Technologies (PoPETs), 2026.</i>
FAccT '26	CounterFace: A Synthetic Face Dataset for Fine-Grained Counterfactual Evaluation of Face Recognition Systems [Code] [Paper] Guruprasad V. Ramesh , Harrison Rosenberg, Ashish Hooda, Shimaa Ahmed, Ramya Korlakai Vinayak, Kassem Fawaz. <i>ACM Conference on Fairness, Accountability, and Transparency (FAccT), 2026.</i>
AAAI '24	Limitations of Face Image Generation (Co-first author) [Code] [Paper] [Long Report] Harrison Rosenberg*, Shimaa Ahmed*, Guruprasad V. Ramesh* , Kassem Fawaz, Ramya Korlakai Vinayak. <i>Proceedings of the AAAI Conference on Artificial Intelligence, 2024</i>
ISCA '26	Analyzing gem5 Commit-Code Inconsistency with LLMs Elise Song, Aaryan Patel, Guruprasad V. Ramesh , Jack West, Lance Hartung, Ethan Cecchetti, Kassem Fawaz, Joshua San Miguel, Matthew D. Sinclair. <i>gem5 Workshop at International Symposium on Computer Architecture (ISCA), 2026.</i>
Interspeech '23	Federated Representation Learning for Automatic Speech Recognition [Paper] Guruprasad V. Ramesh , Gopinath Chennupati, Milind Rao, Anit Kumar Sahu, Ariya Rastrow, Jasha Droppo. <i>ISCA SPSC Symposium at Interspeech, 2023.</i>

RESEARCH EXPERIENCE

Apr 2023 – Present	Graduate Research Assistant , <i>Wisconsin Privacy and Security Group</i> University of Wisconsin – Madison · Advisor: Prof. Kassem Fawaz - Currently working on designing and evaluating web agents for user privacy tasks on websites. - Built synthetic data generation pipelines, including synthetic website creation for UI element detection and benchmarks for browser-use agents on web privacy and security tasks.
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- Researched fairness in human face generation with diffusion models; audited face recognition systems and evaluated face understanding in diffusion models.

Jun 2025 – Aug 2025 **Research Intern, Otter AI**

Mentor: Kritika Singh

- Led research on contextual deep biasing for ASR in Otter's production speech system.
- Benchmarked and integrated open-source OCR models on internal datasets, enabling contextual handling of technical vocabulary and a 15% reduction in Word Error Rate.
- Implemented enhancements that improved general WER by 2% while preserving technical vocabulary accuracy.

Jun 2022 – Dec 2022 **Applied Scientist Intern, Amazon Alexa**

Mentor: Dr. Gopinath Chennupati

- Conducted research on representation learning for ASR with Alexa Speech, resulting in a paper at the ISCA SPSC Symposium in Interspeech 2023.
- Designed experiments and simulated datasets; integrated production-level code with robust unit testing.

Aug 2020 – Jun 2021 **ML Project Assistant, IIT Madras Speech Lab**

Advisor: Prof. Umesh Srinivasan

- Researched low-resource multilingual ASR systems for Indian languages, achieving a 1–2% WER improvement with wav2vec 2.0-based models over Transformer models.
- Deployed ASR models for English and Indian languages in an Indian government-funded speech to speech translation project.
- Co-organized English and Hindi ASR Challenges with 20+ participants: developed baseline models, verified test data, and managed the leaderboard.

AWARDS & HONORS

- 2026 Travel Grant, Proceedings on Privacy Enhancing Technologies Symposium (PETS).
- 2026 Travel Grant, ACM Conference on Fairness, Accountability, and Transparency (FAccT).
- 2025 Distinguished Artifact Reviewer Award, PETS.
- 2024 Wisconsin Alumni Research Foundation (WARF) Accelerator Grant.

TALKS

- Apr 2026 *Evaluating Web Agents on Website Security and Privacy Tasks.*
Contextual Agent Privacy & Security Talks, Google Research.
- Oct 2024 *Using Large Language Models (LLMs) in Research.*
Computer Engineering Seminar, UW – Madison ECE Department.
- Aug 2023 *Federated Representation Learning for ASR.*
ISCA SPSC Symposium at Interspeech 2023.

ACADEMIC SERVICE

Reviewer: ICLR 2025, CVPR 2025, IEEE TASLP Journal 2025.

Artifact Evaluation: PETS 2025 (*Distinguished Reviewer Recognition*), USENIX Security 2026.

External Reviewer: WACV 2026, USENIX Security Symposium 2025, IEEE S&P 2024–2026.

TEACHING

2023 – 2026 **Teaching Assistant**, ECE 532: *Matrix Methods in Machine Learning*
University of Wisconsin – Madison

- Organized weekly office hours, discussion sessions, and in-class problem-solving activities for 600+ students.
- Designed quiz questions and monthly assessments.

SKILLS

Programming Languages: Python, Bash, $\text{\LaTeX}$, SQL.

ML Frameworks & Libraries: PyTorch, Vllm, Transformers, NumPy, Pandas, Scikit-learn, Matplotlib, PySpark, ESPnet, Kaldi, Fairseq.